Permanent Hardness by Multilinearization

An algebraic descendant of Aaronson's linear-optical proof

Abstract

We give a self-contained algebraic proof that exact computation of the permanent of an integer matrix is #P-hard. The proof is inspired by Aaronson's linear-optical proof, but its formal language is only commuting polynomials and linear substitutions. Starting from #CircuitSAT, we view a computation as a choice of one local truth-table row at every gate, subject only to pairwise consistency constraints. A forbidden pair xy is converted by the polarization identity

$$xy \longmapsto (x+y)(x-y) = x^2 - y^2$$

into non-multilinear terms, which are then discarded. Multilinear projection itself is compiled into an ordinary linear substitution using one fresh marker variable. After all markers are deferred to the final coefficient extraction, the entire reduction is one linear substitution; the relevant multilinear coefficient is a permanent.

The guiding picture is: *logical conflicts are turned into routing collisions, and the permanent sums collision-free routings.*

The proof in one line.

$$\begin{aligned}\text{\#CircuitSAT} &\longrightarrow \text{local choices with forbidden pairs}\\ &\longrightarrow \text{conflict } xy \mapsto x^2 - y^2 \longrightarrow \text{multilinear projection} \longrightarrow \mathrm{Per}\,.\end{aligned}$$

The first arrow is parsimonious; the middle step is polarization; the last step is the elementary identity saying that a multilinear coefficient of a linear substitution is a permanent.

1 Why the permanent is a multilinear routing coefficient

Let $z_1, \dots, z_M$ be commuting variables. For an $M \times M$ matrix $A = (A_{ij})$ over a field of characteristic zero, define the linear substitution T_A by

$$T_A(z_j) = \sum_{i=1}^{M} A_{ij} z_i,$$

and extend it linearly and multiplicatively to all polynomials. Thus

$$T_A \circ T_B = T_{AB}.$$

For a set $J \subseteq [M]$, write

$$z_J := \prod_{j \in J} z_j.$$

A monomial is *multilinear* if every variable has exponent at most one. Write $\mathrm{ML}(f)$ for the *multilinear part* of a polynomial f: we simply delete every monomial in which some variable occurs with exponent at least two.

The permanent identity we need is almost tautological.

Lemma 1.1 (Permanent as the multilinear part)**.** *If $|J| = n$, then*

$$\mathrm{ML}\big(T_A(z_J)\big) = \sum_{\substack{I\subseteq[M]\\|I|=n}} \mathrm{Per}(A_{I,J})\, z_I.$$

Equivalently, for $|I| = |J|$,

$$[z_I]\, T_A(z_J) = \mathrm{Per}(A_{I,J}).$$

Proof. Expand

$$T_A(z_J) = \prod_{j\in J}\left(\sum_i A_{ij} z_i\right).$$

A term in the expansion chooses one destination i for each $j \in J$. The term is multilinear exactly when these destinations are all distinct. To obtain z_I, the choices must therefore form a bijection $J \to I$. Summing the weights of all such bijections gives

$$\sum_{\sigma: J\xrightarrow{\sim} I} \prod_{j\in J} A_{\sigma(j),j} = \mathrm{Per}(A_{I,J}). \qquad \square$$

Routing picture. Think of the factors indexed by J as anonymous tokens. Each token independently chooses an output mode according to the corresponding column of A. A repeated variable z_i^2 means that two tokens collided at output i. The multilinear part keeps exactly the collision-free routings. The permanent is their total weight.
This is also the bosonic identity underlying Aaronson's proof: commuting variables make every permutation contribute with the same sign. We will use the routing picture as intuition, but no quantum formalism is needed.

So the permanent already knows how to do one useful thing: it sums independent choices while enforcing a *no-collision* condition. Our reduction will arrange that every logically inconsistent pair of choices produces a collision.

2 A manifestly #P-complete local-choice problem

We start from #CircuitSAT: given a Boolean circuit C with free input bits, count the inputs on which C outputs one. This is a direct complete problem for #P: a polynomial-time verifier can be compiled into a polynomial-size Boolean circuit, with its witness bits as the free inputs.

We next rewrite a circuit computation in a form tailored to pairwise consistency.

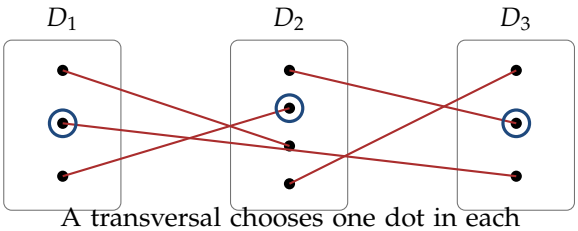
2.1 Conflict-free transversals

Definition 2.1. A *conflict system* consists of finite sets (or "boxes")

$$D_1, \ldots, D_m$$

and a set of conflict edges between elements belonging to different boxes. A *transversal* chooses one element $a_u \in D_u$ from each box. It is *conflict-free* if no two chosen elements form a conflict edge.

The associated counting problem asks for the number of conflict-free transversals.



A transversal chooses one dot in each box. It is valid precisely when no red conflict edge joins two chosen dots.

Lemma 2.2 (CircuitSAT becomes local consistency)**.** *There is a parsimonious polynomial-time reduction from* #CircuitSAT *to counting conflict-free transversals in which every box has size at most four.*

Proof. Assume for convenience that C is a fan-in-two NAND circuit. (Any fixed universal Boolean basis would work.)

For each input node u, create a box

$$D_u = \{0, 1\}.$$

The chosen element is the value assigned to that circuit input.

For each NAND gate g, create a box

$$D_g = \{00, 01, 10, 11\}.$$

Choosing $ab \in D_g$ means that the two inputs of g have values a, b. Its output value is then already determined:

$$\mathrm{out}_g(ab) = \mathrm{NAND}(a, b).$$

Thus every element of D_g is itself a valid local truth-table row; there is no separate gate constraint to enforce.

Now consider a wire from a node u to, say, the first input of a gate g. Put a conflict edge between local states $a \in D_u$ and $b \in D_g$ exactly when they disagree about the value on that wire:

$$\mathrm{out}_u(a) \neq \mathrm{in}_{g,1}(b).$$

Do the analogous thing for every wire. Finally, if g_{out} is the output gate, delete from its box the unique row 11, on which NAND outputs zero. Equivalently, its box is $\{00, 01, 10\}$.

Given an input assignment, deterministic circuit evaluation determines exactly one local state at every gate, and those states form a conflict-free transversal exactly when the circuit accepts. Conversely, in any conflict-free transversal, wire consistency forces each gate's chosen row to be the row induced by the values of its predecessors; by induction through the circuit, the transversal is exactly the evaluation of its chosen input assignment. The output restriction forces acceptance.

Hence satisfying inputs and conflict-free transversals are in bijection. □

The circuit has now disappeared. From this point onward, the task is simply:

> For each box, choose one local state. Count the choices that avoid a given collection of forbidden pairs.

3 The ideal algebra: turn conflicts into non-multilinear terms

Introduce one commuting variable z_a for every local state a in every box. If the boxes are $D_1, \dots, D_m$, then

$$F_0 := \prod_{u=1}^{m} \left(\sum_{a \in D_u} z_a \right)$$

is the generating polynomial of all transversals: every monomial of F_0 chooses exactly one state from each box, and every such monomial has coefficient one. In particular, F_0 is multilinear.

Suppose now that x and y are two variables corresponding to a forbidden pair of local choices. We want an operator that acts on a multilinear polynomial by deleting exactly the monomials containing xy.

3.1 Polarization: a mixed term becomes a collision

Consider the invertible two-variable change of coordinates

$$B: \qquad x \longmapsto x + y, \qquad y \longmapsto x - y.$$

Its inverse is

$$B^{-1}: \qquad x \longmapsto \frac{x+y}{2}, \qquad y \longmapsto \frac{x-y}{2}.$$

The key identity is simply

$$\boxed{xy \overset{B}{\longmapsto} (x+y)(x-y) = x^2 - y^2.}$$

Thus the bad mixed term xy is converted into a sum of non-multilinear terms.

Let $\mathrm{ML}_{x,y}$ mean: keep only those terms in which both x and y have degree at most one, leaving all other variables untouched.

Lemma 3.1 (Conflict deletion by polarization)**.** *On polynomials that are multilinear in x, y, the operator*

$$\mathcal{D}_{x,y} := T_{B^{-1}} \circ \mathrm{ML}_{x,y} \circ T_B$$

deletes exactly those monomials divisible by xy, and leaves every other monomial unchanged.

Proof. It suffices to check the four-dimensional space spanned by

$$1, \quad x, \quad y, \quad xy.$$

On $1, x, y$, applying B produces a polynomial of degree at most one in each variable, so $\mathrm{ML}_{x,y}$ does nothing, and B^{-1} restores the original polynomial. On the bad term,

$$xy \overset{B}{\mapsto} x^2 - y^2 \overset{\mathrm{ML}_{x,y}}{\longmapsto} 0.$$

Thus

$$1 \mapsto 1, \qquad x \mapsto x, \qquad y \mapsto y, \qquad xy \mapsto 0. \qquad \square$$

This is the conceptual core of the reduction. A logical conflict is a mixed monomial xy. Polarization turns it into a collision $x^2 - y^2$, and taking the multilinear part rejects collisions.

> **Bosonic mnemonic.** The identity $xy \mapsto x^2 - y^2$ says that one token in each of two modes is transformed into a superposition of two tokens occupying the same mode. In linear optics this is "bunching." Here it is just the polarization identity for the quadratic form xy.

Because $\mathcal{D}_{x,y}$ is, on multilinear polynomials, literally the diagonal operation "delete monomials containing xy," all such conflict filters commute, even when conflicts share variables.

Proposition 3.2 (The ideal computation)**.** *If E is the set of conflict pairs, then*

$$F_* := \left(\prod_{\{x,y\} \in E} \mathcal{D}_{x,y} \right) F_0$$

is exactly

$$F_* = \sum_{\tau \text{ conflict-free transversal}} z_\tau, \qquad z_\tau := \prod_{a \in \tau} z_a.$$

Proof. Every monomial of F_0 is a transversal. For each conflict $\{x, y\}$, the operator $\mathcal{D}_{x,y}$ deletes precisely the transversal monomials that choose both endpoints. Thus a monomial survives all filters exactly when the corresponding transversal contains no conflict edge. Surviving coefficients remain one. $\square$

At this stage the counting reduction is conceptually complete. We have used only linear substitutions and the operation "take the multilinear part." What remains is to remove that one apparently nonlinear operation.

4 Multilinear projection from a private marker

The next identity is the main technical simplification over a more literal linear-optical presentation.

For one variable x, write

$$f = f_0 + f_1 x + f_2 x^2 + \cdots,$$

where none of the coefficients f_k involves x. Define

$$\mathrm{ML}_x(f) := f_0 + f_1 x,$$

the part of f having x-degree at most one.

Introduce a fresh variable r, which we call a *marker*.

Lemma 4.1 (Marker identity)**.** *For every polynomial f independent of r,*

$$\boxed{\mathrm{ML}_x(f) = [r]\, f(r)(x + r).}$$

Equivalently, let $L_{x,r}$ be the ordinary linear substitution

$$L_{x,r} : \qquad x \longmapsto r, \qquad r \longmapsto x + r.$$

Then

$$\boxed{[r]\, T_{L_{x,r}}(f\, r) = \mathrm{ML}_x(f).}$$

Proof. We have

$$f(r)(x + r) = (f_0 + f_1 r + f_2 r^2 + \cdots)(x + r).$$

The coefficient of r^1 is exactly

$$f_0 + f_1 x.$$

The second formulation is the same identity, since

$$T_{L_{x,r}}(f(x)r) = f(r)(x + r).$$ □

There is no cancellation hidden in this identity.

Overflow picture. Suppose the exponent of x records how many tokens occupy a mode, and the fresh marker r initially contains one token. Under

$$x \mapsto r, \qquad r \mapsto x + r,$$

all tokens formerly in x are sent to the marker location, while the marker token may either stay there or move into x. At the end, demand exactly one token in r.
If x initially had zero tokens, the marker stays. If x had one token, that token occupies r while the marker moves to x. If x had two or more tokens, at least two tokens necessarily occupy r, so the requested final monomial is impossible. The marker is therefore a one-slot *overflow detector*.

Since

$$\mathrm{ML}_{x,y} = \mathrm{ML}_x \,\mathrm{ML}_y,$$

we can implement the multilinear projection in each conflict filter using two fresh markers, one for x and one for y.

4.1 Why all marker tests may be deferred

The marker lemma appears to extract a coefficient immediately after the substitution. To end with a single permanent, we want to postpone every such extraction until the very end. Freshness makes this automatic.

Lemma 4.2 (Deferred markers)**.** *Let r be a variable, and let S be any linear substitution that does not involve r: it fixes r and no other variable is mapped to an expression containing r. Then*

$$[r]\,T_S(f) = T_S\big([r]\,f\big).$$

Consequently, if each marker introduced by theorem 4.1 is fresh and is never used again, all marker coefficient extractions can be deferred to the end of the computation.

Proof. Write $f = \sum_k f_k r^k$, with each f_k independent of r. Since S does not involve r,

$$T_S(f) = \sum_k T_S(f_k) r^k.$$

Taking the coefficient of r gives $T_S(f_1)$. Iterating this observation handles any number of fresh markers. □

Thus a conflict filter can be compiled as the following sequence of *ordinary linear substitutions*:

$$\boxed{B \quad\longrightarrow\quad L_{x,r_x} \quad\longrightarrow\quad L_{y,r_y} \quad\longrightarrow\quad B^{-1},}$$

where r_x, r_y are private markers that begin and end with exponent one. By theorems 4.1 and 4.2, the final coefficient in those markers is exactly the ideal operator $\mathcal{D}_{x,y}$.

5 Assembling the reduction into one permanent

We now combine the pieces. Let

$$C = (D_1, \ldots, D_m; E)$$

be the conflict system obtained from a circuit by theorem 2.2. For every local state $a \in D_u$, we already have a state variable z_a. Introduce additionally one *root variable* q_u for each box.

For each conflict edge $e = \{x, y\} \in E$, introduce two private marker variables $r_{e,x}, r_{e,y}$.

Let

$$R := \prod_{e=\{x,y\}\in E} r_{e,x} r_{e,y} \qquad \text{and} \qquad Q := \prod_{u=1}^{m} q_u.$$

Our initial and final monomial will be the same multilinear monomial

$$M := QR.$$

Step 1: spread each local choice

For each box D_u, apply the linear substitution

$$q_u \longmapsto \sum_{a \in D_u} z_a.$$

Applied to Q, this produces exactly

$$F_0 = \prod_u \left(\sum_{a \in D_u} z_a \right).$$

The markers in R are untouched.

Step 2: delete all conflicts

For every conflict $e = \{x, y\}$, apply its compiled filter

$$B_{x,y}, \qquad L_{x,r_{e,x}}, \qquad L_{y,r_{e,y}}, \qquad B_{x,y}^{-1}.$$

Never touch the two markers again.

By the marker and deferral lemmas, extracting the coefficient of all the marker variables at the end has exactly the same effect as applying the ideal conflict deletion operators. Hence, after all conflict filters, the relevant marker coefficient is

$$F_* = \sum_{\tau \text{ conflict-free transversal}} z_\tau.$$

Step 3: gather each box

Finally, for every $a \in D_u$, apply

$$z_a \longmapsto q_u.$$

Every transversal monomial z_τ contains exactly one state variable from each box, so it maps to Q. Therefore

$$[M] \text{ final polynomial} = \#\{\text{conflict-free transversals of } C\}.$$

By theorem 2.2, this is exactly the number of satisfying inputs of the original circuit.

But every operation we have just listed is an ordinary linear substitution. Compose them all into one matrix A. Let I be the set of coordinates appearing in M, namely the root modes together with all marker modes. Then

$$M = z_I,$$

and by theorem 1.1,

$$[z_I]\, T_A(z_I) = \mathrm{Per}(A_{I,I}).$$

Set $P_C := A_{I,I}$. The number of modes and elementary substitutions is polynomial in the circuit size, so P_C can be computed explicitly in polynomial time with polynomial-bit rational entries. We have proved:

Theorem 5.1 (Permanent hardness over the rationals)**.** *Given a Boolean circuit C, one can construct in polynomial time a rational matrix P_C of polynomial dimension such that*

$$\boxed{\operatorname{Per}(P_C) = \#\{x : C(x) = 1\}.}$$

Notice what has happened. The multilinear projections were useful for *understanding* the computation, but after introducing private markers they vanished from the final construction. The actual reduction is a product of ordinary matrices, and the permanent identity is applied only once, at the very end.

6 Clearing denominators

The only nonintegral coefficients in the construction come from

$$B^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

There are only polynomially many conflict filters. Hence the entries of P_C are rationals with polynomial bit complexity; for example, multiplying by $D = 2^t$, where t is the number of conflict filters, clears all denominators.

Let N be the dimension of P_C, and set

$$\widehat{P} := DP_C.$$

Then $\widehat{P}$ is an integer $N \times N$ matrix and, by homogeneity of the permanent,

$$\operatorname{Per}(\widehat{P}) = D^N \operatorname{Per}(P_C) = D^N \#\{x : C(x) = 1\}.$$

The known factor D^N can be divided out in polynomial time. We therefore obtain the main theorem.

Theorem 6.1. *Exact computation of the permanent of an integer matrix is #P-hard.*

The construction uses signed integer matrices. Valiant's classical theorem gives the stronger standard statement that the permanent of a 0-1 matrix is #P-complete [2]. The point of the present proof is the hardness mechanism, not the subsequent deweighting to 0-1 entries.

7 What the proof is really doing

It is useful to separate the four conceptual moves from the bookkeeping that implements them.

1. **Computation is local choice plus consistency.** A circuit evaluation is a choice of one valid truth-table row at every gate. The only global requirement is that neighboring rows agree on the wires between them.
2. **Consistency violations are forbidden pairs.** Once local truth-table rows are the states, every wire condition is simply a list of pairs of local states that may not be chosen together.

3. **A forbidden pair becomes a collision.** If x, y represent two incompatible states, polarization gives
$$xy \mapsto x^2 - y^2.$$
Thus the logically bad configuration is moved from the multilinear sector into the non-multilinear sector.

4. **The permanent sums collision-free routings.** A multilinear coefficient of a linear substitution is a permanent. Private markers let every intermediate multilinear projection be postponed until that single final coefficient extraction.

The reduction can therefore be remembered by one slogan:

Turn logical conflicts into routing collisions.

The marker construction is not a second conceptual mechanism. It is merely the compiler that lets us express the ideal operation "discard non-multilinear terms" using ordinary linear substitutions. Its identity
$$\mathrm{ML}_x(f) = [r]\, f(r)(x + r)$$
can be viewed as a one-slot capacity test: occupancy zero or one fits, occupancy two or more overflows.

8 Relation to Aaronson's linear-optical proof

Aaronson gave a striking proof of permanent hardness using linear-optical quantum computation and the Knill–Laflamme–Milburn universality construction [1]. Two features of that proof survive here in purely algebraic form. First, the bosonic transition-amplitude identity becomes theorem 1.1: a multilinear coefficient of a linear substitution is a permanent. Second, bosonic bunching becomes the polarization identity
$$xy \mapsto x^2 - y^2.$$

What disappears is universality. Instead of dynamically simulating an arbitrary circuit, we first rewrite the circuit as a static local-consistency problem. Then the only interaction required is "forbid this pair," which polarization plus multilinear projection implements directly. Thus the quantum language remains useful as historical and pictorial motivation, but the formal proof is entirely a statement about commuting variables and their multilinear part.

References

[1] Scott Aaronson, *A Linear-Optical Proof that the Permanent is #P-Hard*, Proceedings of the Royal Society A **467** (2011), 3393–3405.

[2] Leslie G. Valiant, *The Complexity of Computing the Permanent*, Theoretical Computer Science **8** (1979), 189–201.